CS-302: Artificial Intelligence

LECTURE NO 1:

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Organization of Lecture

1. Introduction of Artificial Intelligence(AI)
2. Reasons of Boost in AI
3. Goals of AI
4. Applications of AI
5. Composition of AI
6. Advantages and Disadvantages of AI
7. Classifications of AI

1. Introduction of Artificial Intelligence (AI)

Artificial Intelligence (A I)

↳ A I is the study of how to make computer do things which people do better [machine + human Intelligence]

→ A I can cause a machine to work as human.

→ A I
- → Artificial [Man-Made] → Not made by nature
- → Intelligence [Power of thinking] → given by human

2. Reasons of Boost in AI

Reasons of Boost in AI

i) S/w or device can be made to solve Real-time problems.

ii) Creation of virtual assistant. [SIRI, CORTANA].

iii) Robots development.
↳ [Helps in dangerous enviornment conditions, For example diffusing bomb by sending robot rather than sending a human to diffuse bomb]

iv) New Job opportunities in field of AI with every invention of new techology.

3. Goals of AI

Goals of AI

i) Replication of human intelligence i.e., infusing human intelligence into a machine.

ii) Solving problems that require knowledge/Previous knowledge.

iii) Building a machine that can do human intelligence tasks.

↳ [Playing CHESS, Proving Theorems, automated car driven.....]

iv) Providing advise to the user (using virtual assistant).

v) Intelligent connection b/w perception & action.

4. Applications of AI (1/4)

① AI in Gaming:

- Game b/w human and machine.
- Machine can think large no. of moves
- For example: Chess, Poker, tic-tac-toe etc.

② AI in Natural Language Processing (NLP):

- Machine can understand human Language
- For example virtual assistant like SIRI, CORTANA etc.

③ AI in Health care:

- Fast diagnosis with the help of AI.
- Robotic Surgery, i.e., Any surgeon from pakistan can perform surgery in USA with the help of certain robots.

4. Applications of AI (2/4)

④ AI in Finance:

↳ Adaptive Intelligence (ability of the mind to be able to change in response to the current demand in the enviornmant).

↳ Adaptive Intelligence helps in making automatic chatbot, algorithm trading i.e An algorithm tells which share should be bought & which not be bought

⑤ AI in Data Security:

↳ Helps in make data/application more secure

↳ Examples are AEG bot (Automatic Exploit Generation) & AI2 platform.. They can detect bugs in any s/w and secures it automatically.

4. Applications of AI (3/4)

⑥ Expert Systems:

↳ Integration of S/W, machine and special information to provide reasoning and advise.

↳ An expert system is a computer program that is designed to solve complex problems & to provide decision-making ability like a human do.

⑦ Computer Vision:

↳ Understand the visual automatically by machine.

⑧ Speech Recognition:

↳ Extract the meaning of sentence by human talk.

↳ For example: slang removal, noise removal etc.

4. Applications of AI (4/4)

(9) **Robotics:**

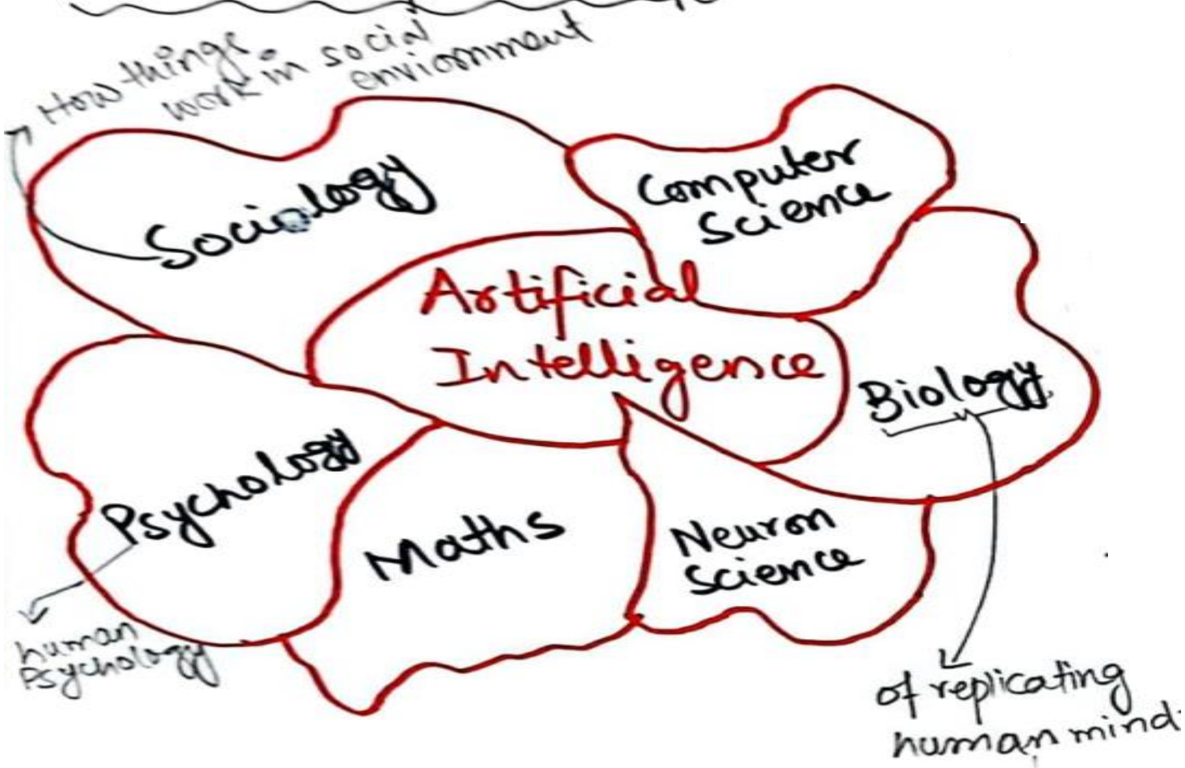
- A.I has a remarkable role in robotics.
- Most intelligent humanoid robots are Erica & Sophia.
  - They can talk and behave like humans.

(10) **AI in Electronic-Commerce (E-commerce):**

- Automatic Recommendation of product: Recommending user a product by understanding his choices & patterns.
- Handling service request (chatbots in e-commerce are all A I based automatic systems).

5. Composition of AI

AI is comprised of :-



All above elements of AI are combination of :

→ Reasoning
→ Learning
→ Probelm Solving
→ Language Understanding

6. Advantages and Disadvantages of AI

Advantages of AI	Disadvantages of AI
Accuracy ↑ & error ↓	Cost ↑
High Speed (Fast Decision Making).	Can't think beyond the limits it can only perform in a domain related to its knowledge.
Reliability is more	No feelings & emotions. (A robot can't be replaced by a human nurse).
Usefulness in Risky Area	More dependency on machines.
Digital/Virtual Assistant	No original thinking (It has built-in knowledge).

7. Classifications of AI (1/3)

Classification of AI:

Weak AI:

- Also called Narrow AI.
- Weak AI is able to perform dedicated task with intelligence [Not concerned how the tasks are performed, but it is concerned with performance & efficiency].
- Can't perform beyond its field or limitations.
- Examples:
  - > Flying machine
  - > using logics
  - > Apple SIRI
  - > Play chess.

7. Classifications of AI (2/3)

Strong AI:

↳ Also called General AI.

↳ It is the study and design of machines that simulate human mind to perform intelligent tasks.

↳ i) Borrowing ideas from psychology & neuroscience. (to replicate human intelligence).

↳ ii) Forgetting thing, Genetics, language understanding. (weaknesses of human).

7. Classifications of AI (3/3)

Evolutionary AI:

↳ It is the study and design of machines simulate simple creatures and attempt to evolve.

↳ Examples Ants, bees etc. (Many algorithms are specifically designed on ants & bees).

Super AI

↳ You can say it is a hypothetical concept for now

↳ Machines would be made that are more better than humans in terms of thinking and all human aspects..

↳ For example:

[machine $\xleftrightarrow{\text{Communication}}$ machine]

References: Textbooks

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5. *"AI: A new Synthesis", by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *"Artificial Intelligence: Theory and Practice" by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*